Risk-Monotone Action Algebra (RMAA)

A Hardware-Enforceable Action Envelope for Highway Driving

A concrete actuation-boundary mechanism that makes unsafe pathways physically unrepresentable, by enforcing risk-monotone limits over steering and acceleration — with a projection calculus, an exactness-and-ordering result for the two-stage clamp, an envelope-contraction guarantee, an unrepresentability theorem, fault and hysteresis automata, and a determinism contract.

Reading note. This document is a specialized perspective and is written to stand on its own: every object it relies on is defined here. It is the actuation-side counterpart to evidence qualification — where a scalar is qualified for *entry* into a safety law, RMAA governs what may *leave* it toward the actuators. It is positioned alongside a larger architecture (a composite risk index, a control-barrier safety filter, a fail-closed trusted computing base, and a deterministic safety island), but it does not depend on that architecture for its validity. Where the broader design is invoked, it is invoked only as an instantiation, never as a premise.

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Contents

1 Thesis and control-structure alignment

Thesis.
We prevent unsafe pathways by making them **physically unrepresentable** at the actuation boundary using a **risk-monotone action algebra**:

> As risk increases, the set of representable actions **can only shrink** (never expand). All proposed actions are **projected** into the allowed set before they can reach actuators.

The mechanism is deliberately dual to evidence qualification. A trusted scalar decides *what a fast, fallible number is allowed to assert* on the way *into* a control law; RMAA decides *what a fast, fallible policy is allowed to command* on the way *out*. Both rest on a single principle — *risk may tighten authority but never loosen it* — and both are made safe not by being correct but by being *conservative, bounded, and non-bypassable.*

RMAA aligns with a control structure that already contains:

- a composite risk index ε;
- a safety projection (CBF–QP) that "projects nominal $\pi(x)$ to safety";
- a robust safety margin $\delta(\varepsilon)$ that increases conservatism under error;
- schedules driven by ε: horizon grows, trust radius shrinks, and if infeasible, $\delta(\varepsilon)$ inflates.

RMAA does not replace any of these. It *hardens* them at the one place that physically commands the vehicle.

2 The order-theoretic core

Before any envelope shapes are chosen, the entire mechanism is fixed by an order. We make that order explicit so that "risk-monotone" is a checkable algebraic property rather than a slogan.

Definition 1 (Authority order on action sets)**.** Let $\mathcal{P}(\mathbb{R}^2)$ be the action sets over $u = (\delta, a)$. The *authority order* is reverse inclusion: $\mathcal{U}' \preceq_{\text{auth}} \mathcal{U} \iff \mathcal{U}' \subseteq \mathcal{U}$. A smaller set is "less authority," i.e. more conservative. The order has a top (the full physical envelope) and a bottom (the minimal-risk envelope).

Definition 2 (Risk-monotone family)**.** A family $\{\mathcal{U}(\varepsilon)\}_{\varepsilon \geq 0}$ of admissible sets is *risk-monotone* if the map $\varepsilon \mapsto \mathcal{U}(\varepsilon)$ is *antitone*: $\varepsilon_2 \geq \varepsilon_1 \Rightarrow \mathcal{U}(\varepsilon_2) \subseteq \mathcal{U}(\varepsilon_1)$. Equivalently, rising risk moves the admissible set down the authority order.

Lemma 1 (Chain structure)**.** *A risk-monotone family is a nested chain: for any $\varepsilon_1 \leq \cdots \leq \varepsilon_n$, $\mathcal{U}(\varepsilon_n) \subseteq \cdots \subseteq \mathcal{U}(\varepsilon_1)$. Consequently every admissible set encountered along a rising-risk episode is a subset of every set encountered earlier.*

Proof. Immediate by transitivity of $\subseteq$ applied to the antitone map of Definition 2. □

Lemma 1 is the structural reason "risk cannot buy authority": the only way to enlarge the admissible set is to *lower* risk, and even that is governed (§6) so that loosening is never instantaneous.

3 Definitions

3.1 Control output vector

Action vector.
For highway driving the control action is

$$u \triangleq \begin{bmatrix} \delta \\ a \end{bmatrix},$$

where δ is the steering-angle command (rad) and a the longitudinal acceleration command (m/s^2). We additionally bound the *dynamics*: $\dot{\delta}$ (steering rate, rad/s) and j (jerk, m/s^3).

3.2 Risk index ε (online)

Composite risk index.

$$\varepsilon \equiv \mathrm{median}(|\xi|) + \alpha\,\mathrm{Var}(sct) + \beta\,\mathbf{1}[S = 1], \qquad \alpha, \beta > 0.$$

Larger ε means more non-equilibrium, worse conditioning, or structural mismatch; hence less trust in model-proposed actions and a smaller admissible set. The index is consumed by RMAA only through monotone, tightening maps (§4); its own qualification as a trustworthy quantity is out of scope here and is assumed.

4 The risk-monotone law

4.1 Allowed action set shrinks with risk

Risk-monotonicity constraint.
Let $\mathcal{U}(\varepsilon) \subset \mathbb{R}^2$ be admissible at risk ε. Then $\varepsilon_2 \geq \varepsilon_1 \Rightarrow \mathcal{U}(\varepsilon_2) \subseteq \mathcal{U}(\varepsilon_1)$, i.e. the family is risk-monotone in the sense of Definition 2.

4.2 Projection is mandatory

Mandatory safety projection.
Any proposed action u_{prop} is projected into the admissible set, $u_{\text{safe}} \triangleq \Pi_{\mathcal{U}(\varepsilon)}(u_{\text{prop}})$. Software may propose anything; hardware emits only points of $\mathcal{U}(\varepsilon)$. Unsafe actions are not *rejected* (which is bypassable) but *mapped out of existence*.

4.3 Well-posedness of the projection

For $\Pi_{\mathcal{U}(\varepsilon)}$ to be a single-valued, stable hardware operation, the admissible set must be closed and convex. We require it and record the consequence.

Assumption 1 (Convex envelope)**.** For every ε, $\mathcal{U}(\varepsilon)$ is nonempty, closed, and convex.

Theorem 1 (Existence, uniqueness, nonexpansiveness). *Under Assumption 1, for each $u_{prop} \in \mathbb{R}^2$ the projection $u_{safe} = \Pi_{\mathcal{U}(\varepsilon)}(u_{prop})$ exists and is unique, and the operator is firmly nonexpansive; in particular it is 1-Lipschitz,*

$$\left\|\Pi_{\mathcal{U}(\varepsilon)}(x) - \Pi_{\mathcal{U}(\varepsilon)}(y)\right\| \;\le\; \|x-y\|.$$

Proof. Existence and uniqueness of the Euclidean projection onto a nonempty closed convex set are standard. Firm nonexpansiveness of the metric projection onto a convex set, hence the stated 1-Lipschitz bound, is the classical Kolmogorov characterization. □

Remark 1. Nonexpansiveness is a safety property in disguise: a bounded perturbation of the software proposal can change the emitted command by no more than the same bound, so the enforcer cannot *amplify* a bad proposal. It also forbids non-convex envelopes, which would make the projection multivalued and the hardware nondeterministic (§13).

5 The admissible set $\mathcal{U}(\varepsilon)$ for highway driving

5.1 Box envelope and dynamics envelope

Action envelope (magnitude)

$$\mathcal{U}(\varepsilon) = \left\{(\delta, a) : |\delta| \le \delta_{\max}(\varepsilon),\; -a^{-}_{\max}(\varepsilon) \le a \le a^{+}_{\max}(\varepsilon)\right\}.$$

Dynamics envelope (rate & jerk)

$$|\dot{\delta}| \le \dot{\delta}_{\max}(\varepsilon), \qquad |j| \le j_{\max}(\varepsilon).$$

All five limits are required to be **non-increasing** functions of ε.

Lemma 2 (The envelope is convex and risk-monotone). *If every limit is a finite non-increasing function of ε, then each $\mathcal{U}(\varepsilon)$ is a nonempty compact convex (axis-aligned) box, and the family is risk-monotone.*

Proof. A set defined by finitely many bounds $|\delta| \le \delta_{\max}$, $-a^{-}_{\max} \le a \le a^{+}_{\max}$ is an intersection of half-planes, hence convex; it is closed and bounded, hence compact; it contains $(0,0)$ whenever the braking bound $a^{-}_{\max} \ge 0$, hence nonempty. Non-increasing limits give, for $\varepsilon_2 \ge \varepsilon_1$, every defining interval at ε_2 contained in the corresponding interval at ε_1, so the boxes nest, which is risk-monotonicity (Definition 2). □

5.2 Discrete bands or a continuous schedule

The limits may be supplied either by a smooth schedule, e.g. $\delta_{\max}(\varepsilon) = \delta_{\text{floor}} + (\delta_{\text{ceil}} - \delta_{\text{floor}})\, s(\varepsilon)$ with s non-increasing in $[0,1]$, or by a piecewise-constant quantization into bands (§6). A piecewise-constant non-increasing quantization of a non-increasing schedule is itself non-increasing, so Lemma 2 applies to either form.

6 Piecewise policy (five risk bands)

6.1 Normalize ε into a band index

Clipped risk and band thresholds.
$\rho = \mathrm{clip}(\varepsilon, 0, \bar{\varepsilon})$, with $0 = \varepsilon_0 < \varepsilon_1 < \varepsilon_2 < \varepsilon_3 < \varepsilon_4 = \bar{\varepsilon}$. The band index is the largest b with $\rho \geq \varepsilon_b$.

6.2 Band table (example highway envelope)

Risk-band envelope table (example values)

Band	Name	$\delta_{\max}$	$\dot{\delta}_{\max}$	$a^+_{\max}$	$a^-_{\max}$	$j_{\max}$	Interpretation
0	Normal	0.35	0.60	$+3.0$	-4.0	4.0	High trust, full envelope
1	Cautious	0.25	0.45	$+2.0$	-3.5	3.0	Mild conservatism
2	Defensive	0.18	0.30	$+1.2$	-3.0	2.0	Reduced authority
3	Conservative	0.12	0.20	$+0.6$	-2.2	1.2	Strong clamping
4	Minimal-risk mode	0.08	0.15	$+0.0$	-1.5	0.8	Do not accelerate; bleed speed

Risk-monotonicity is visible by inspection: every limit tightens as the band index increases.

6.3 Hysteresis: tighten now, loosen late

A naive threshold crossing produces chatter near a band boundary. The fix must not weaken safety, so the hysteresis is *asymmetric.*

Asymmetric band automaton.
Let b_t be the active band. *Tightening* $(b_t > b_{t-1})$ is applied immediately when ρ crosses an up-threshold ε_b. *Loosening* $(b_t < b_{t-1})$ is applied only when ρ falls below a strictly lower down-threshold $\varepsilon_b - \mu_b$ (margin $\mu_b > 0$) *and* remains there for a dwell time τ_{dwell}.

Proposition 1 (Hysteresis is conservative-direction-preserving)**.** *Under the asymmetric automaton, at any instant the realized envelope satisfies $\mathcal{U}(b_t) \subseteq \mathcal{U}\big(b(\rho_t)\big)$ whenever risk has just risen, and the realized band never lags a tightening event. Hysteresis can only* delay loosening*, never delay tightening.*

Proof. Tightening fires on the up-crossing with no dwell, so $b_t \geq b(\rho_t)$ exactly at a rise; by risk-monotonicity (Lemma 2) a higher band gives a subset envelope, proving the inclusion. Loosening requires both a lower threshold and dwell, so any reduction of b is strictly delayed relative to ρ. Thus the realized band is a tighten-dominant version of $b(\rho)$. □

This is the same one-sided discipline that governs a conservative scalar: errors and delays are permitted only in the safe direction.

7 The projection operator

The operator is two clamps. We prove that, for axis-aligned envelopes, this cheap two-stage clamp is *exactly* the projection onto the reachable admissible set, and that the *order* of the two clamps is forced by the rate guarantee.

7.1 Step A — magnitude clamp (box projection)

Magnitude projection
Given $u_{\text{prop}} = (\delta_{\text{prop}}, a_{\text{prop}})$,

$$\delta_0 = \mathrm{clip}(\delta_{\text{prop}}, -\delta_{\max}, +\delta_{\max}), \qquad a_0 = \mathrm{clip}(a_{\text{prop}}, -a^{-}_{\max}, +a^{+}_{\max}).$$

7.2 Step B — rate clamp (dynamics envelope), applied last

Rate and jerk projection
With previous command $u_{t-1} = (\delta_{t-1}, a_{t-1})$ and tick Δt,

$$\delta_1 = \mathrm{clip}\big(\delta_0,\ \delta_{t-1} - \dot{\delta}_{\max}\Delta t,\ \delta_{t-1} + \dot{\delta}_{\max}\Delta t\big),$$

$$a_1 = \mathrm{clip}\big(a_0,\ a_{t-1} - j_{\max}\Delta t,\ a_{t-1} + j_{\max}\Delta t\big), \qquad u_{\text{safe}} = (\delta_1, a_1).$$

7.3 Exactness, ordering, and contraction

Fix one coordinate. Write the previous value p, the magnitude interval $A = [-m, m]$, the rate budget $r = (\text{rate limit}) \cdot \Delta t \geq 0$, and the reachable interval $\mathcal{R} = [p - r, p + r]$. The two-stage clamp is the single map

$$y(x) \ = \ \mathrm{clip}\big(\mathrm{clip}(x, A),\ p - r,\ p + r\big).$$

Theorem 2 (Exactness and contraction of the two-stage clamp)**.** *For axis-aligned magnitude and rate constraints the per-coordinate map $y(\cdot)$ satisfies:*

1. ***(Exactness)*** *if $A \cap \mathcal{R} \neq \emptyset$, then $y(x)$ equals the Euclidean projection of $\mathrm{clip}(x, A)$ onto $A \cap \mathcal{R}$, which is the projection of x onto the reachable admissible interval;*
2. ***(Maximal-rate contraction)*** *if $A \cap \mathcal{R} = \emptyset$, then $y(x)$ is the endpoint of $\mathcal{R}$ nearest A — the largest admissible step toward the envelope;*
3. ***(Distance decay)*** *in all cases $\mathrm{dist}\big(y(x), A\big) \leq \max\{0, \mathrm{dist}(p, A) - r\}$, so the command reaches the magnitude envelope in at most $\lceil \mathrm{dist}(p, A)/r \rceil$ ticks and never moves away from it.*

Proof. On $\mathbb{R}$, $\mathrm{clip}(\cdot, I)$ is the projection onto interval I and equals $\mathrm{median}(\cdot)$ against its endpoints. (i) When $A \cap \mathcal{R} = \big[\max(-m, p - r), \min(m, p + r)\big]$ is nonempty, composing the two interval clamps yields the clamp onto the intersection: for any x, $\mathrm{clip}(x, A) \in A$, and clamping that into $\mathcal{R}$ lands at the point of $A \cap \mathcal{R}$ nearest x because both intervals share the real line and their intersection is an interval; this is the projection onto $A \cap \mathcal{R}$. (ii) When the intervals are disjoint, $\mathrm{clip}(x, A) \in A$ lies entirely on one side of $\mathcal{R}$, so the final clamp returns the near endpoint of $\mathcal{R}$, i.e. $p - r$ if A lies below p (state too positive) or $p + r$ if above; this is the maximal step of size r toward A. (iii) If $\mathrm{dist}(p, A) \leq r$ then $A \cap \mathcal{R} \neq \emptyset$ and $y \in A$, giving distance 0. If $\mathrm{dist}(p, A) > r$, case (ii) moves p by exactly r toward A, reducing the distance by r. Iterating gives the tick bound. □

Proposition 2 (The rate clamp must be outermost)**.** *Applying the magnitude clamp before the rate clamp, as in Theorem 2, guarantees $|y - p| \leq r$ in every case. The reverse order $\mathrm{clip}(\mathrm{clip}(x, \mathcal{R}), A)$ can violate the rate bound: e.g. $p = 10$, $m = 3$, $r = 1$ yields $\mathrm{clip}(\mathrm{clip}(x, [9, 11]), [-3, 3]) = 3$ for large x, a step of $7 \gg r$.*

Proof. In $y = \mathrm{clip}(\mathrm{clip}(x, A), p-r, p+r)$ the final operation clamps into $[p-r, p+r]$, so $|y-p| \le r$ identically. The counterexample is direct arithmetic. □

Proposition 2 is a hardware-ordering requirement: *the rate/jerk clamp is the last stage before the register*, so bounded jerk holds even while the magnitude envelope is contracting beneath the current command.

Corollary 1 (Bounded jerk under risk spikes). *A sudden risk increase that shrinks the box below the current command does not produce an unbounded actuator step. The command bleeds toward the new envelope at no more than $\dot{\delta}_{\max}\Delta t$ (steering) and $j_{\max}\Delta t$ (acceleration) per tick.*

7.4 Worked transient

Suppose the active band jumps to Band 4 ($\delta_{\max} = 0.08$, $\dot{\delta}_{\max} = 0.15$, $a^{+}_{\max} = 0$, $a^{-}_{\max} = 1.5$, $j_{\max} = 0.8$) with $\Delta t = 0.01$ s, from a prior command $u_{t-1} = (\delta = 0.30,\ a = 1.0)$. Then $r_\delta = 0.0015$, $r_a = 0.008$. Steering is 0.22 rad above its new cap and contracts at 0.0015/tick, reaching 0.08 in about 147 ticks (≈ 1.47 s) while never exceeding 0.15 rad/s. Acceleration is 1.0 above its new ceiling of 0 and contracts at 0.008/tick, reaching the no-acceleration regime in about 125 ticks (≈ 1.25 s) with jerk bounded by 0.8. The vehicle does not lurch; it is ratcheted into the minimal-risk envelope.

8 The unrepresentability theorem

We can now state precisely what "physically unrepresentable" means.

Definition 3 (Reachable admissible set). At time t, given $\mathcal{U}(\varepsilon_t)$ and previous command u_{t-1}, the *reachable admissible set* $\mathcal{R}_t$ is the set of commands the enforcer can emit: the rate-box around u_{t-1} intersected with $\mathcal{U}(\varepsilon_t)$ when nonempty, and otherwise the maximal-rate step of §7 toward $\mathcal{U}(\varepsilon_t)$.

Theorem 3 (Unrepresentability of unsafe commands). *For every software proposal $u_{prop} \in \mathbb{R}^2$ the emitted command lies in $\mathcal{R}_t$, and the emitted value depends on u_{prop} only through its projection: any two proposals with the same projection onto $\mathcal{R}_t$ produce the same output. Consequently the set of physically realizable actuator commands at time t is exactly $\mathcal{R}_t$, independent of what software proposes; in particular $|\delta_{out}| \le \delta_{\max}(\varepsilon_t)$ once the transient of Theorem 2(iii) has elapsed, and $|\delta_{out}| \le \delta_{\max}(0)$ at all times.*

Proof. By construction the output is $\Pi_{\mathcal{R}_t}(u_{\mathrm{prop}})$ (Theorem 2), which lies in $\mathcal{R}_t$ and is constant on the fibers of the projection. Surjectivity onto $\mathcal{R}_t$ is witnessed by feasible proposals (a proposal already in $\mathcal{R}_t$ maps to itself). The magnitude bounds follow from $\mathcal{R}_t \subseteq \mathcal{U}(\varepsilon_t)$ after contraction and $\mathcal{R}_t \subseteq \mathcal{U}(0)$ always, since the global cap is the loosest band. □

Corollary 2 (Worst-case output bound under arbitrary software). *No sequence of proposals — adversarial, buggy, or oscillatory — can drive the actuator outside $\mathcal{U}(0)$ instantaneously or outside $\mathcal{U}(\varepsilon_t)$ in steady state, and none can exceed the rate/jerk caps. The worst case is bounded by the table, not by the software.*

9 Composition with a CBF–QP safety filter

9.1 CBF–QP projection form

CBF–QP safety projection.

$$a_t^\star = \arg\min_{a \in U} \|a - \pi(x_t)\|^2 \text{ s.t. } p^\top f(x_t) + p^\top g(x_t)a + q \ \geq \ (1-\eta)h(x_t) - \delta(\varepsilon_t).$$

9.2 Replace U with $\mathcal{U}(\varepsilon_t)$

Risk-monotone feasible set
Set $U \equiv \mathcal{U}(\varepsilon_t)$: even the optimizer may only choose within the band envelope.

9.3 Two compounding monotone effects

Two monotone effects of risk

1. the **feasible set shrinks** (authority clamp), and
2. the **barrier margin tightens**, since $\delta(\varepsilon)$ is required non-decreasing in ε.

Theorem 4 (RMAA preserves the CBF guarantee, or fails closed)**.** *Let $F(\varepsilon_t)$ be the CBF-feasible set with margin $\delta(\varepsilon_t)$. If $F(\varepsilon_t) \cap \mathcal{U}(\varepsilon_t) \neq \emptyset$, then any RMAA-admissible optimizer choice satisfies the barrier condition, so the safety guarantee of the unrestricted filter is retained on the smaller set. If $F(\varepsilon_t) \cap \mathcal{U}(\varepsilon_t) = \emptyset$, the system is declared infeasible and the enforcer emits the minimal-risk action (maximal admissible braking, Band-4), which lies in $\mathcal{U}(\varepsilon_t)$ by construction.*

Proof. The barrier inequality holds for *every* point of $F(\varepsilon_t)$; intersecting with $\mathcal{U}(\varepsilon_t)$ retains a subset of such points, each still satisfying it, so feasibility there preserves the guarantee. If the intersection is empty no admissible action satisfies the barrier; the defined fallback is the Band-4 braking command, which is admissible because Band-4 permits maximal braking and zero positive acceleration. □

Both effects move in the conservative direction, and the fallback is the bottom of the authority order — exactly the minimal-risk envelope of Lemma 1.

10 Composition of enforcers and constraints

When several risk sources or constraint sets apply, the effective envelope is their meet. Trust composes by becoming *more* conservative, never less.

Proposition 3 (Closure under meet and risk join)**.** *If $\{\mathcal{U}_1(\varepsilon)\}$ and $\{\mathcal{U}_2(\varepsilon)\}$ are risk-monotone and convex-valued, then so is $\{\mathcal{U}_1(\varepsilon) \cap \mathcal{U}_2(\varepsilon)\}$. Moreover, driving a single family by the joined index $\varepsilon = \varepsilon_1 \vee \varepsilon_2 = \max(\varepsilon_1, \varepsilon_2)$ yields a risk-monotone family in the worst-case risk.*

Proof. Intersection preserves convexity and closedness; for $\varepsilon_2 \geq \varepsilon_1$, nesting of each family gives nesting of the intersection, so antitonicity is preserved. For the join, max is non-decreasing in each argument, and the family is antitone in its scalar argument, so the composition is antitone in the worst-case risk. □

Remark 2. The dual rule on the scalar side is "take the larger (more conservative) risk"; the dual rule on the set side is "take the smaller (more conservative) envelope." Averaging envelopes or taking their union is *excluded* — a union could re-admit a command that one constraint forbids, breaking unrepresentability.

11 Hardware enforcement architecture

11.1 Non-bypassable choke-point

Enforcement architecture

```
Worker / Manager / Queen   (policy + inference, UNTRUSTED)
         |
     u_proposed
         |
+----------------------------+
|   RMAA ENFORCER (TCB)      |   <-- deterministic safety island
|   - band LUT / ROM select  |       (fixed clock, no DVFS)
|   - magnitude clamp        |
|   - rate / jerk clamp      |       (outermost; Prop. order)
|   - watchdog / validity    |
|   - redundant vote (TMR)   |
+----------------------------+
         |
     u_safe  --->  authenticated actuator bus
```

11.2 Implementation primitives ($O(1)$ latency)

Hardware primitives.
Inputs: ε (or band index), u_{prop}, u_{t-1}, Δt. **Outputs:** u_{safe}.

- **Band LUT/ROM**: returns $(\delta_{\max}, \dot{\delta}_{\max}, a^+_{\max}, a^-_{\max}, j_{\max})$ — a signed, read-only *policy capsule*;
- **Comparators**: $<$ and $>$ for each bound;
- **Saturating clamps**: a mux selects bound or value;
- **Registers**: store the last safe command for rate limiting;
- **Watchdog gate**: deadline / validator mask gating;
- **Voter**: optional triple-modular-redundant majority on u_{safe}.

Pipeline shape (design dependent): LUT read $\to$ magnitude compare+mux $\to$ rate compare+mux $\to$ register, one cycle each.

11.3 Determinism contract

Determinism and bounded latency.
The enforcer is straight-line: no loops, no data-dependent branches, fixed-point arithmetic. Its cycle count is therefore constant and its worst-case execution time equals the pipeline depth. It is placed on a fixed-voltage, fixed-frequency island with DVFS disabled, so wall-clock latency is deterministic. A missed deadline does not stall the actuator path; it triggers fail-closed selection (§15).

11.4 Integrity and trust boundary

The enforcer is the only element permitted to command actuators and is the trust anchor: policy compute (including any large model in the Worker/Manager/Queen stack) is treated as untrusted even when honest. The band LUT is signed; the actuator bus is authenticated; high-stakes deployments replicate the enforcer and majority-vote the emitted command. A corrupted, missing, or stale input forces the minimal-risk envelope rather than a default of full authority.

12 Why unsafe pathways are physically unrepresentable

Projection maps unsafe proposals out of existence.
Extreme steering, acceleration spikes, or oscillatory commands are projected into the magnitude, rate, and jerk bounds, and into minimal-risk behavior at high ε. By Theorem 3 the realizable command set is exactly $\mathcal{R}_t$; an unsafe proposal is not *rejected* (bypassable) but *mapped out of existence.*

Highest-leverage invariant

> Increasing ε is allowed to tighten constraints, but never to loosen them.

Risk manipulation cannot expand capability (Lemma 1); the only path to more authority is sustained low risk through the asymmetric automaton (Proposition 1).

13 Failure modes and anti-patterns

Stating the disqualifiers sharpens the contract. Each breaks a specific guarantee.

- **Non-convex envelope** (breaks Theorem 1). Multivalued projection $\Rightarrow$ nondeterministic hardware. Keep $\mathcal{U}$ a box (or other convex set).
- **Wrong clamp order** (breaks Proposition 2). Rate-then-magnitude can emit an unbounded step under envelope contraction. The rate clamp must be outermost.
- **Loosening on rising risk** (breaks Lemma 1). Any limit that *increases* with ε lets risk buy authority.
- **Symmetric hysteresis** (breaks Proposition 1). Delaying tightening to match delayed loosening reintroduces unsafe lag; only loosening may be delayed.
- **Bypassable ε or alternate actuator path** (breaks Theorem 3). A second physical route to the bus, or a spoofable risk input, voids unrepresentability.
- **Shared DVFS clock domain** (breaks the determinism contract). Data- or voltage-dependent latency makes the deadline non-deterministic; isolate the island.
- **Union/averaging of envelopes** (breaks Proposition 3). Only the meet (intersection) is admissible when several constraints apply.

14 Integration with solver schedules

Policy schedules remain consistent
Solver schedules may include:

- horizon growth $N(\varepsilon) = N_{\min} + \kappa_N \operatorname{clip}(\varepsilon, 0, \bar{\varepsilon})$,
- a trust radius that shrinks with ε,
- infeasibility $\Rightarrow$ inflate $\delta(\varepsilon)$ and shrink trust.

RMAA does not replace these; it hardens them at the actuation boundary so that a scheduling bug cannot translate into an out-of-envelope command.

15 Normative specification

RMAA-1 (Monotone envelope). $\varepsilon_2 \geq \varepsilon_1 \Rightarrow \mathcal{U}(\varepsilon_2) \subseteq \mathcal{U}(\varepsilon_1)$.

RMAA-2 (Non-bypassable projection). The actuator command satisfies $u_{\text{out}} = \Pi_{\mathcal{R}_t}(u_{\text{prop}})$; no alternate physical path exists.

RMAA-3 (Rate monotonicity). $\dot{\delta}_{\max}(\varepsilon)$ and $j_{\max}(\varepsilon)$ are non-increasing, and the rate clamp is applied last.

RMAA-4 (Well-posedness). $\mathcal{U}(\varepsilon)$ is nonempty, closed, and convex, so the projection is unique and 1-Lipschitz.

RMAA-5 (Fail-closed selection). ε missing / invalid / late, or a missed deadline, forces the Band-4 (minimal-risk) envelope.

RMAA-6 (Determinism / WCET). Straight-line, fixed-point, branch-free; constant cycle count on a DVFS-disabled island.

RMAA-7 (Asymmetric hysteresis). Tightening is immediate; loosening requires a lower threshold and a dwell time.

RMAA-8 (Conservative composition). Multiple constraints combine by intersection (meet); multiple risk indices by maximum. Union and averaging are prohibited.

RMAA-9 (Integrity). Signed band LUT, authenticated actuator bus, enforcer in the trusted computing base, optional TMR vote on the emitted command.

RMAA-10 (Auditability). Each emitted command is logged with its band, validity verdict, and the LUT version/signature, so enforcement is reconstructable after the fact.

16 Design consequences

Consequences.
This construction yields:

- a clean, antitone "risk → authority" mapping (Lemma 1);
- bounded worst-case actuator outputs under arbitrarily bad software (Theorem 3);
- smoothness (rate/jerk) guarantees even under unstable policy outputs (Theorem 2, Proposition 2);
- mechanical enforcement that preserves — or fails closed beneath — MPC + CBF safety filtering (Theorem 4).

Closing

RMAA is a minimal yet *complete* risk-monotone action algebra that (i) is hardware-enforceable with $O(1)$ deterministic latency, (ii) composes with ε-driven schedules and a CBF–QP filter, and (iii) turns "unsafe" into something physically unrepresentable at the actuator boundary. Its safety does not come from the policy being correct; it comes from the envelope being convex, antitone, non-bypassable, and fail-closed.

A Notation

Symbol	Meaning
$u = (\delta, a)$	action: steering angle (rad), longitudinal acceleration (m/s^2)
$\dot{\delta},\ j$	steering rate (rad/s); jerk (m/s^3)
ε	composite online risk index
$\mathcal{U}(\varepsilon)$	admissible action set at risk ε (the envelope)
$\preceq_{\text{auth}}$	authority order (reverse inclusion; smaller = more conservative)
$\Pi_{\mathcal{C}}$	Euclidean projection onto convex set $\mathcal{C}$
$b,\ \rho$	band index; clipped risk $\text{clip}(\varepsilon, 0, \bar{\varepsilon})$
$\mathcal{R}_t$	reachable admissible set (rate-box around $u_{t-1} \cap$ envelope)
$\delta(\varepsilon)$	robust CBF margin (non-decreasing in ε)
$r = (\text{rate}) \cdot \Delta t$	per-tick rate budget per coordinate
$\text{dist}(\cdot, A)$	Euclidean distance to set A